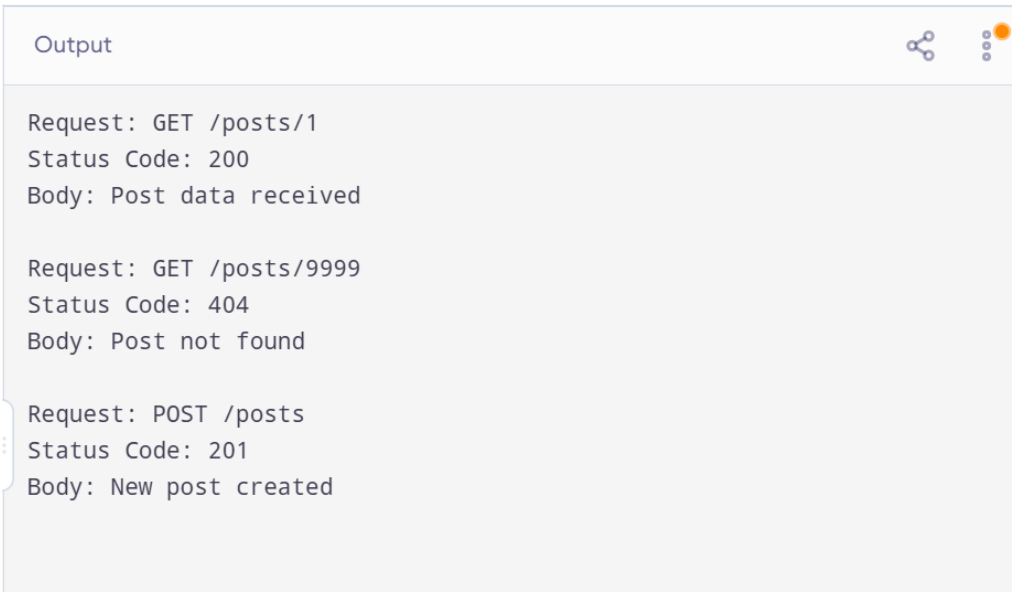
Task 1: HTTP GET and POST Requests — Correct Swift Code

```
import Foundation
func showRequest(_ method: String, _ path: String) {
    print("Request: \(method) \(path)")
    if method == "GET" && path == "/posts/1" {
        print("Status Code: 200")
        print("Body: Post data received")
    } else if method == "GET" && path == "/posts/9999" {
        print("Status Code: 404")
        print("Body: Post not found")
    } else if method == "POST" && path == "/posts" {
        print("Status Code: 201")
        print("Body: New post created")
    }
    print("")
}
showRequest("GET", "/posts/1")
showRequest("GET", "/posts/9999")
showRequest("POST", "/posts")
```



Task 2: HTTP Request Anatomy and Status Codes in Swift

```
import Foundation
let method = "POST"
let path = "/v1/students"
let query = "notify=true"
let version = "HTTP/1.1"
let host = "api.college.example"
let contentType = "application/json"
let authorization = "Bearer abc123"
let body = #"{"name":"Asha","dept":"IT"}"#
print("Method: \(method)")
print("Path: \(path)")
print("Query String: \(query)")
print("HTTP Version: \(version)")
```

```
print("Host: \(host)")
print("Content-Type: \(contentType)")
print("Authorization: \(authorization)")
print("Body: \(body)")
print("200: Success")
print("201: Resource created")
print("204: Success with no response body")
print("400: Bad request")
print("401: Unauthorized")
print("403: Forbidden")
print("404: Resource not found")
print("429: Too many requests")
print("500: Internal server error")
```

Output

```
Method: POST
Path: /v1/students
Query String: notify=true
HTTP Version: HTTP/1.1
Host: api.college.example
Content-Type: application/json
Authorization: Bearer abc123
Body: {"name":"Asha","dept":"IT"}
200: Success
201: Resource created
204: Success with no response body
400: Bad request
401: Unauthorized
403: Forbidden
404: Resource not found
429: Too many requests
500: Internal server error
```

Task 3: Design REST Endpoints for a Student Service

```
import Foundation
struct Student {
    var id: Int
    var name: String
    var department: String
}

var students = [
    Student(id: 1, name: "Asha", department: "IT"),
    Student(id: 2, name: "Ravi", department: "CSE"),
    Student(id: 3, name: "Priya", department: "IT")
]

func handleRequest(_ method: String, _ path: String) {
    if method == "GET" && path == "/students" {
        print("200 OK")
        for student in students {
            print("\(student.id) \(student.name) \(student.department)")
        }
    } else if method == "GET" && path == "/students?department=IT" {
        print("200 OK")
        for student in students where student.department == "IT" {
            print("\(student.id) \(student.name) \(student.department)")
```

```
        }
    } else if method == "GET" && path == "/students/1" {
        if let student = students.first(where: { $0.id == 1 }) {
            print("200 OK: \(student.name), \(student.department)")
        } else {
            print("404 Not Found")
        }
    } else if method == "POST" && path == "/students" {
        students.append(Student(id: 4, name: "Manoj", department: "IT"))
        print("201 Created")
    } else if method == "PUT" && path == "/students/1" {
        if let index = students.firstIndex(where: { $0.id == 1 }) {
            students[index].name = "Asha Kumar"
            students[index].department = "CSE"
            print("200 OK: Student updated")
        }
    } else if method == "DELETE" && path == "/students/3" {
        students.removeAll { $0.id == 3 }
        print("204 No Content")
    } else {
        print("404 Not Found")
    }
}

handleRequest("GET", "/students")
handleRequest("GET", "/students?department=IT")
```

```
handleRequest("GET", "/students/1")
handleRequest("POST", "/students")
handleRequest("PUT", "/students/1")
handleRequest("DELETE", "/students/3")
```

Output

```
200 OK
1 Asha IT
2 Ravi CSE
3 Priya IT
200 OK
1 Asha IT
3 Priya IT
200 OK: Asha, IT
201 Created
200 OK: Student updated
204 No Content
```